COLUMN FOOTING
Queries & Discrepancies Register

Source workbook: **COLUMN FOOTING.xlsx**
Scope: all visible worksheet content; hidden rows/cells excluded from the working scope unless required by an approved visible dependency.

Total items	20	Current status	OPEN - pending engineering decisions
Critical / Critical-query	5	Fault	4
Warning / Query-warning	6	Query / Info	5

Executive observation

The workbook should not yet be used as the direct web-application calculation engine without reconciliation. Two presently important issues are the combined biaxial bearing-pressure condition and the development-length check, both of which can affect the design status. No engineering formula has been silently corrected in this register; the items below are recorded for review and decision.

Detailed discrepancy register

QD-001 - CRITICAL

Cell / Area	E38:E63
Finding	X and Y soil-pressure effects are checked independently. There is no combined biaxial corner-pressure check using P/A +/- Mx/Zx +/- My/Zy.
Engineering impact	Combined factored maximum is approximately 137.61 kN/m2 against 135 kN/m2; working maximum is approximately 91.74 kN/m2 against SBC 90 kN/m2. The combined bearing-pressure condition is therefore potentially UNSAFE even though individual workbook checks show SAFE.
Proposed action	Add live combined biaxial corner-pressure checks, including maximum and minimum pressure, after engineering approval.
Final decision	OPEN - requires approval / clarification

QD-002 - FAULT

Cell / Area	E49, E61
Finding	Working minimum pressures are hard-coded values rather than live formulas.
Engineering impact	They match the present case but can become stale when inputs change, allowing an incorrect status to persist.
Proposed action	Replace hard-coded values with live dependencies linked to the current geometry, load and moment values.
Final decision	OPEN - requires approval / clarification

QD-003 - FAULT

Cell / Area	E117:E123
Finding	Rows labelled required total depth simply copy the required effective depth. Cover and bar allowance are not added.
Engineering impact	Displayed required depths of about 137.47 mm and 150.96 mm should be reconciled with approximately 192.47 mm and 215.96 mm if effective cover is added. The current provided depth of 350 mm still appears adequate.
Proposed action	Separate required effective depth from required overall depth and calculate both explicitly.
Final decision	OPEN - requires approval / clarification

QD-004 - FAULT

Cell / Area	E178:E179
Finding	Minimum-steel references are crossed between X and Y directions: the X calculation uses the Y minimum and the Y calculation uses the X minimum.
Engineering impact	No present numerical effect where flexural Ast governs, but it can produce incorrect reinforcement for other geometries or load cases.
Proposed action	Correct the directional minimum-steel references and verify both axes independently.
Final decision	OPEN - requires approval / clarification

QD-005 - FAULT

Cell / Area	E212, E232
Finding	One-way shear pressure references appear cross-swapped between X and Y directions.
Engineering impact	Reconciled demands are approximately 75.58 kN in Y and 60.89 kN in X instead of the workbook values of about 62.68 kN and 73.42 kN. Both remain SAFE for the present case.
Proposed action	Correct directional pressure references and rerun one-way shear checks.
Final decision	OPEN - requires approval / clarification

QD-006 - CRITICAL

Cell / Area	E195:E196
Finding	Development length required is about 566.4 mm and available length is about 560 mm, but no mandatory SAFE/UNSAFE check is provided.
Engineering impact	The visible design currently has required development length greater than available length, yet the workbook does not flag a failure.
Proposed action	Add a mandatory development-length check to the overall status logic.
Final decision	OPEN - requires approval / clarification

QD-007 - WARNING

Cell / Area	E195
Finding	Development-length bond stress contains hard-coded factors 1.2 x 1.6 rather than a fully linked material/code basis.
Engineering impact	Changing concrete grade may not correctly update the development-length provision.
Proposed action	Move bond-stress provisions into the design-code basis and link them to the selected concrete and steel conditions.
Final decision	OPEN - requires approval / clarification

QD-008 - CRITICAL

Cell / Area	E242:E245
Finding	Concrete bearing enhancement factor is limited using 8. This appears inconsistent with the IS 456 bearing enhancement limit where sqrt(A1/A2) is capped at 2.
Engineering impact	The workbook permissible bearing stress is about 58.62 N/mm2. Using a multiplier cap of 2 would give approximately 27.0 N/mm2. The present demand may still pass, but the existing formula can be materially unconservative.
Proposed action	Verify the intended IS 456 clause and, if approved, cap the enhancement multiplier at the code-compliant limit.
Final decision	OPEN - requires approval / clarification

QD-009 - CRITICAL / QUERY

Cell / Area	E22, E38, E48, E236
Finding	Load-factor basis is internally ambiguous. Several formulas appear to treat the load as factored while E236 multiplies total load by another 1.5.
Engineering impact	A double application or omission of load factor could materially affect footing pressure and column-footing bearing checks.
Proposed action	Confirm whether E9 represents service load or ultimate load, then standardize the load basis throughout the engine.
Final decision	OPEN - requires approval / clarification

QD-010 - QUERY

Cell / Area	E18
Finding	Footing self-weight is approximated as 10% of the design load.
Engineering impact	The pressure calculation does not directly use actual footing dimensions, depth and RCC unit weight.
Proposed action	Confirm whether the 10% allowance is an intentional project assumption and whether SBC is net or gross.
Final decision	OPEN - requires approval / clarification

QD-011 - CRITICAL FOR FUTURE CASES

Cell / Area	Bearing checks
Finding	There is no explicit pmin >= 0 check, eccentricity/no-tension check, or partial-contact logic.
Engineering impact	The present combined minimum pressure remains positive, but larger moments could invalidate the full-contact bearing equations without being detected.
Proposed action	Add no-tension and eccentricity checks and define how partial-contact cases are to be handled.
Final decision	OPEN - requires approval / clarification

QD-012 - WARNING

Cell / Area	E16
Finding	Ru.max = 3.98 is hard-coded even though fck and fy are editable.
Engineering impact	Changing material grades can leave required-depth calculations based on a stale design coefficient.
Proposed action	Derive Ru.max from the selected code/material combination or lock the associated material inputs if it is intentionally fixed.
Final decision	OPEN - requires approval / clarification

QD-013 - QUERY / WARNING

Cell / Area	E173:E175
Finding	Minimum reinforcement uses 0.85bd/fy with a 480 mm section width. The code basis and footing/slab interpretation require confirmation.
Engineering impact	Minimum reinforcement can differ materially depending on whether slab-type minimum steel provisions govern.
Proposed action	Confirm the intended design-code clause and footing strip width convention before final implementation.
Final decision	OPEN - requires approval / clarification

QD-014 - WARNING

Cell / Area	E169:E188
Finding	Required and provided reinforcement are calculated, but there is no explicit mandatory Ast provided >= Ast required status and no complete bar-spacing compliance check.
Engineering impact	The current provided steel exceeds the displayed requirement, but future cases could fail without being reflected in overall status.
Proposed action	Add explicit reinforcement adequacy and spacing checks to the mandatory status set.
Final decision	OPEN - requires approval / clarification

QD-015 - QUERY

Cell / Area	E183, E187
Finding	Bar spacing is calculated as dimension / number of bars + bar diameter.
Engineering impact	This does not directly represent typical centre-to-centre spacing based on N-1 spaces and available width.
Proposed action	Confirm the detailing convention, side cover basis and whether spacing is clear or centre-to-centre.
Final decision	OPEN - requires approval / clarification

QD-016 - WARNING

Cell / Area	E28:E32
Finding	The workbook calculates a required width based on equal projections, but the provided width is forced equal to the provided length.
Engineering impact	This is acceptable for the current 380 x 380 mm column but does not generalize correctly to a rectangular column or unequal footing projections.
Proposed action	Separate provided footing length and width unless the project intentionally restricts the footing to a square plan.
Final decision	OPEN - requires approval / clarification

QD-017 - WARNING

Cell / Area	Workbook metadata
Finding	The workbook contains many legacy external and broken defined names that do not feed the visible calculation flow.
Engineering impact	They add noise and traceability risk but do not need to be reproduced in the Streamlit application.
Proposed action	Exclude unused legacy names from the app scope and document that exclusion.
Final decision	OPEN - requires approval / clarification

QD-018 - QUERY

Cell / Area	E163
Finding	Punching-shear demand is derived using a maximum soil-pressure value and outside area rather than a clearly reconciled axial-load minus reaction-inside-critical-perimeter approach.
Engineering impact	The governing punching-shear demand may depend on the pressure distribution and chosen code formulation.
Proposed action	Reconcile the punching-shear force derivation against the intended code method before coding.
Final decision	OPEN - requires approval / clarification

QD-019 - QUERY

Cell / Area	E76, E82
Finding	Cells labelled pressure at the face of column directly copy the maximum edge pressures.
Engineering impact	For a linear soil-pressure distribution, pressure at the column face normally differs from the edge maximum. The workbook may be using a conservative simplification.
Proposed action	Confirm whether this is deliberate conservatism or a copied-value error.
Final decision	OPEN - requires approval / clarification

QD-020 - INFO

Cell / Area	E17
Finding	pt.min is visible but has no value and does not participate in the current calculation.
Engineering impact	It is a dead or incomplete parameter and should not automatically become an app input.
Proposed action	Leave it out of the application unless its intended purpose is later clarified.
Final decision	OPEN - requires approval / clarification

Recommended review sequence

1. Resolve load basis and load-factor interpretation (QD-009).

2. Confirm combined bearing-pressure and no-tension methodology (QD-001, QD-011).

3. Resolve direct calculation faults affecting live dependencies (QD-002 to QD-005).

4. Resolve development-length and bearing-capacity code provisions (QD-006 to QD-008).

5. Confirm minimum reinforcement, spacing, footing geometry and punching-shear conventions (QD-010, QD-012 to QD-019).

6. Exclude unused metadata and dead inputs from the web application unless later approved (QD-017, QD-020).

Status note

This document is an audit/discrepancy register, not a final design certificate. All entries remain OPEN until the relevant engineering decision is approved. After approval, each correction should be incorporated into the single-source calculation engine and verified by Excel-versus-Web parallel calculation testing.